

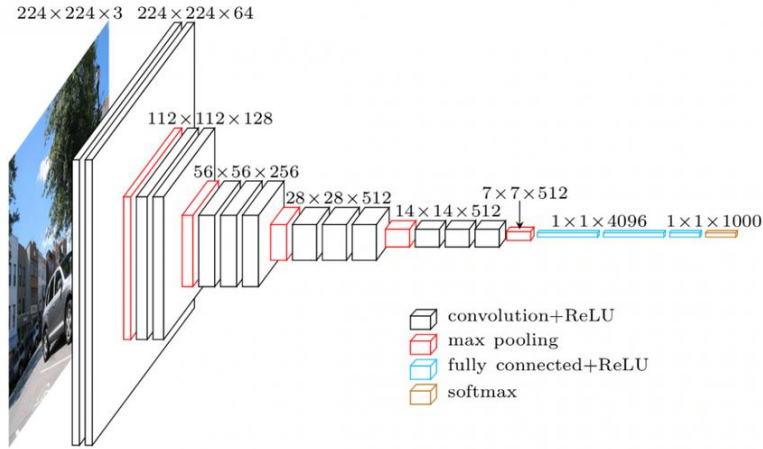


Lecture 3:

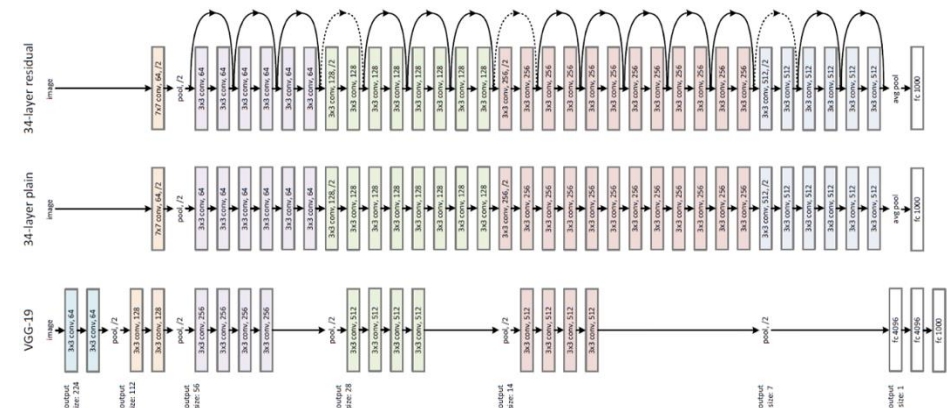
Multimodal Representations

Email: idanschwartz@gmail.com or
idan.schwartz@biu.ac.il

Recall, Deep CNNs for image classification



VGG (2014)



ResNet (2015)

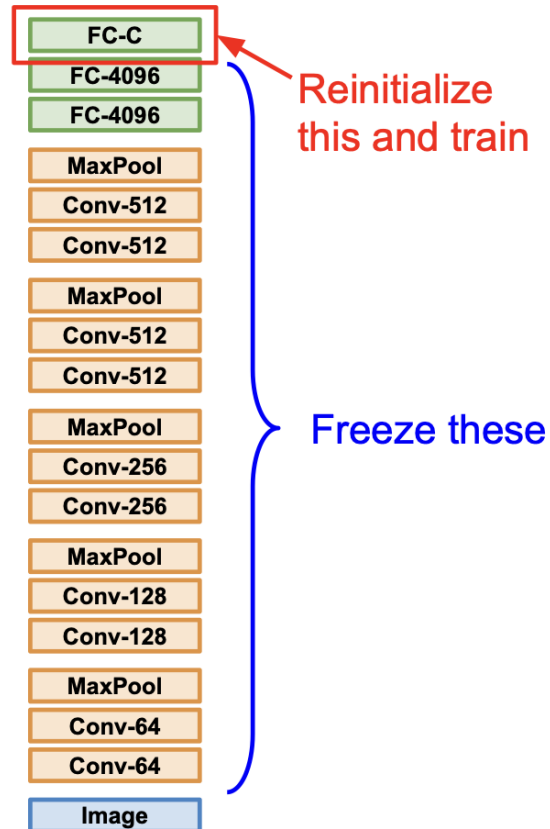
The new tool - Transfer Learning

SOTA in Image representation

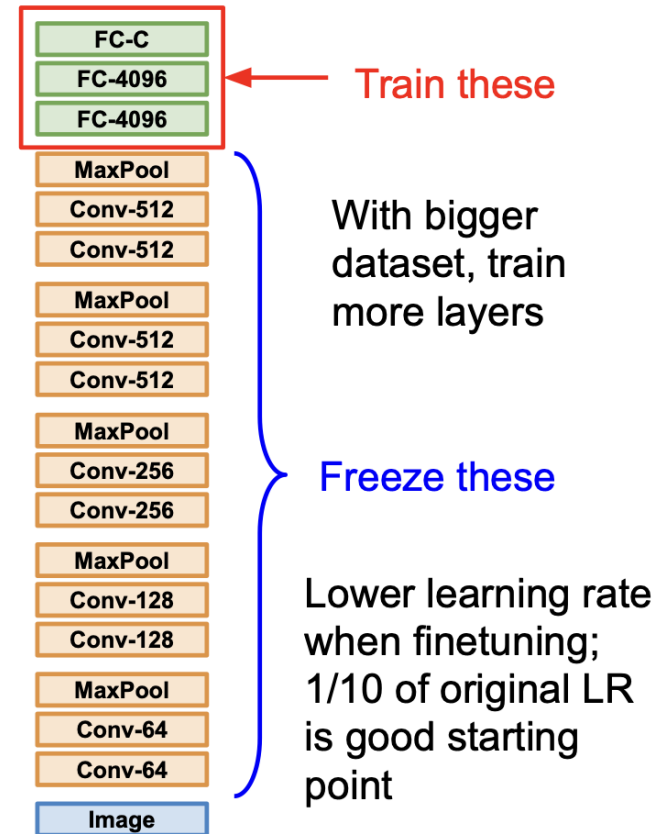
1. Train on Imagenet



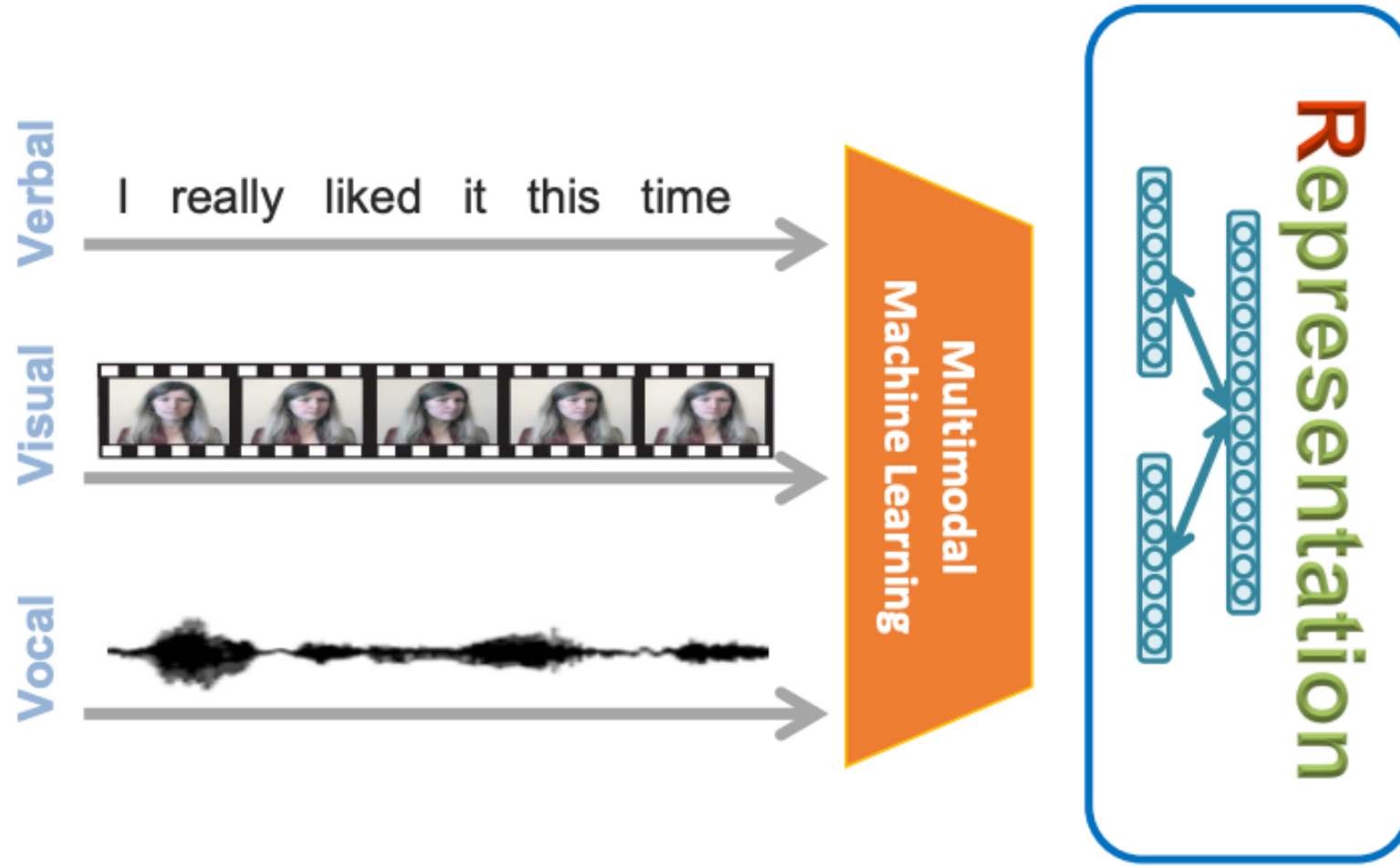
2. Small Dataset (C classes)



3. Bigger dataset



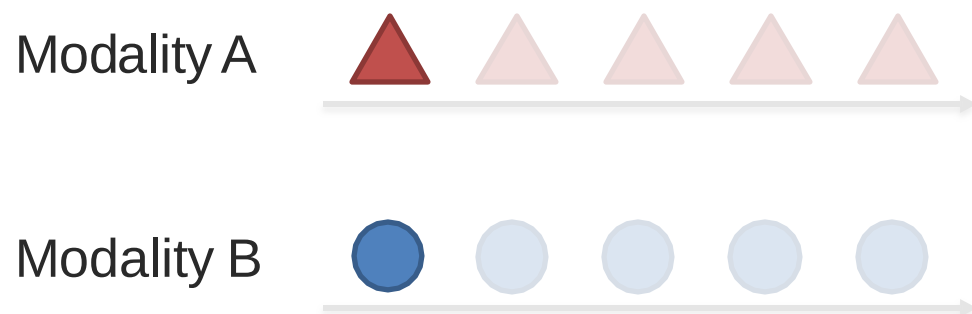
Multimodal Representation



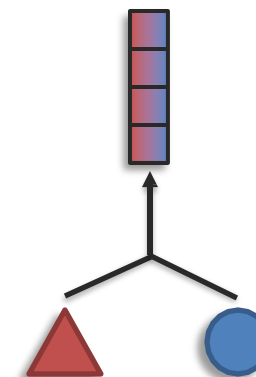
Challenge 1: Representation

Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Individual elements:

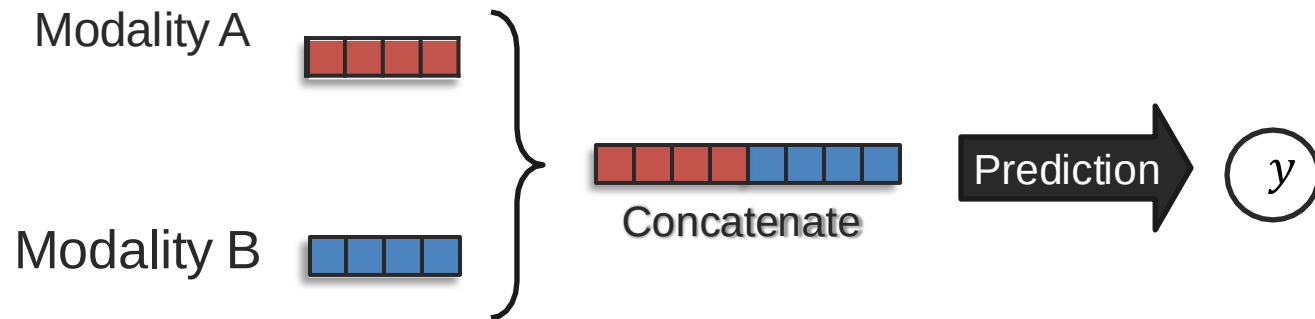


Sub-challenge: Fusion

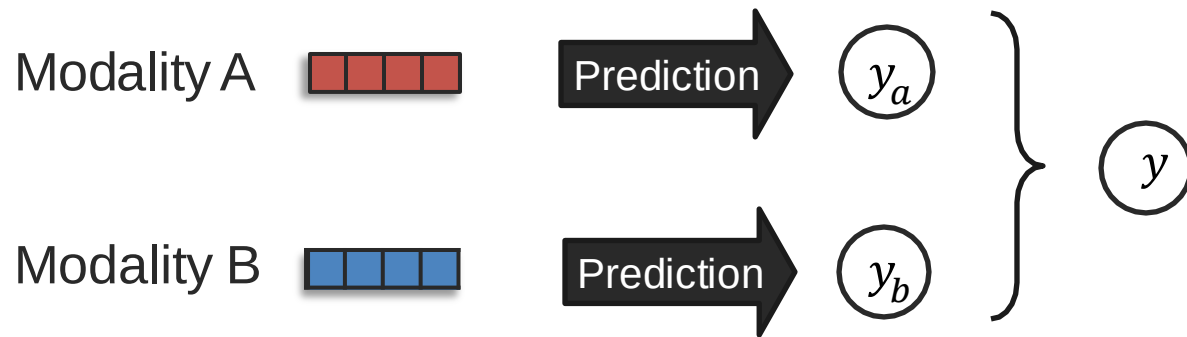


Early and Late Fusion – A historical view

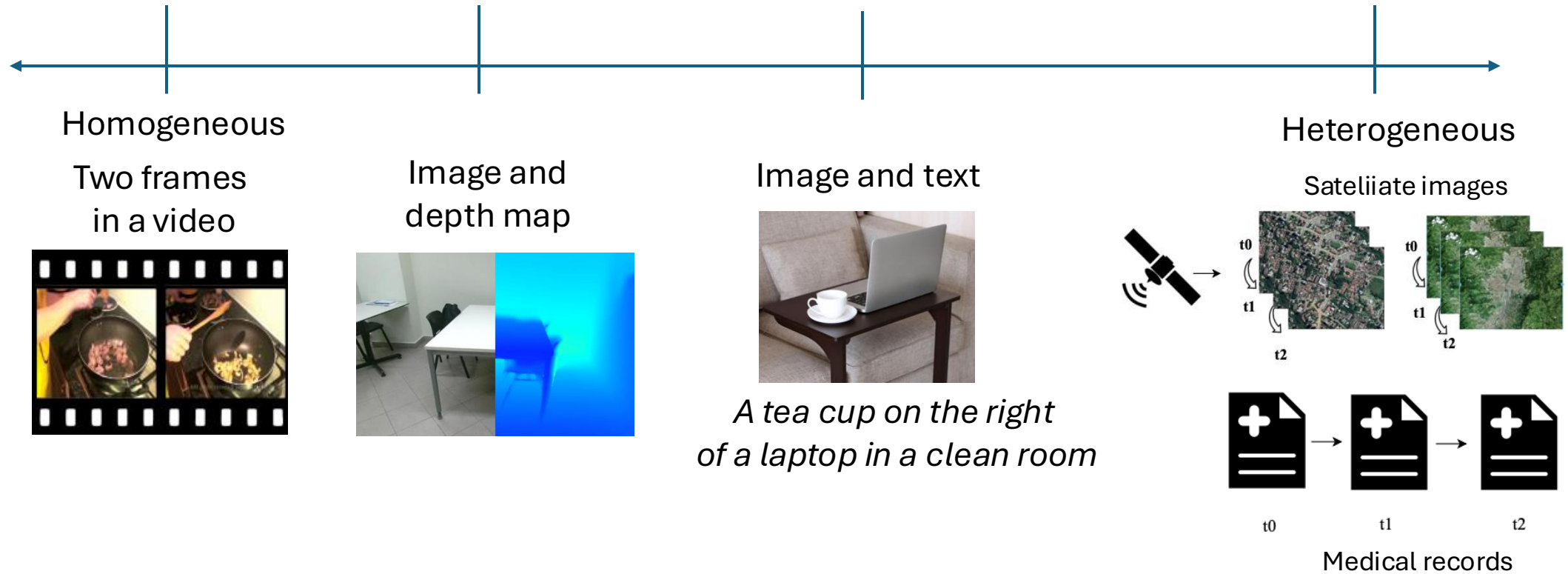
Early fusion:



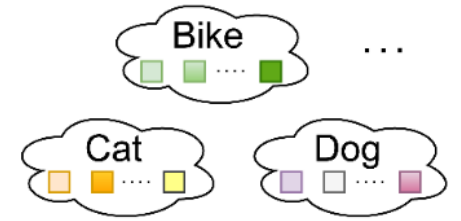
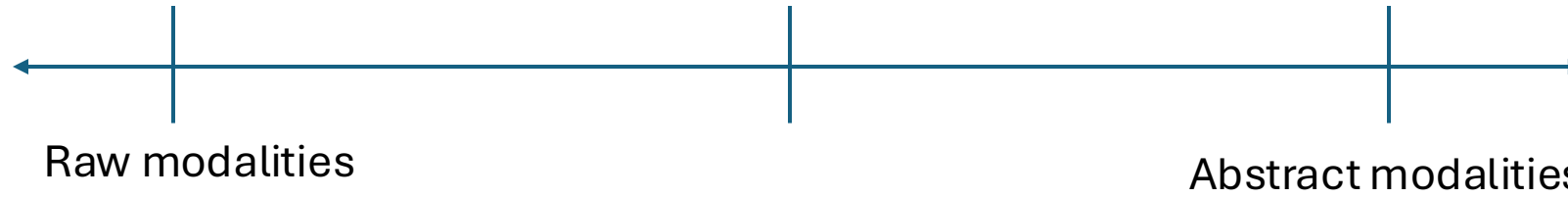
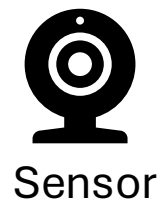
Late fusion (ensemble):



Recall: Modality heterogeneous level



Recall: Modality abstraction level



Examples:

Speech signal

Words

RGB pixels

Objects

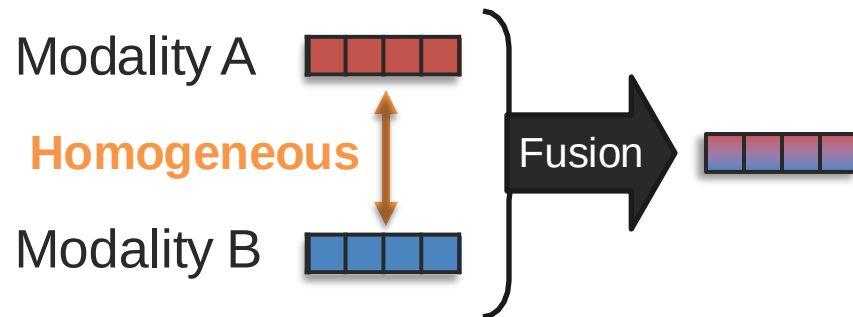
Semantic

Representations

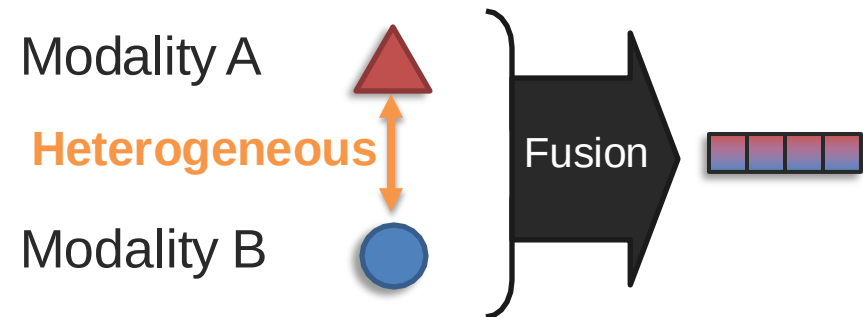
(Usually more homogeneous)

Fusion difficulty - intuition

Basic fusion:



Raw-modality fusion:



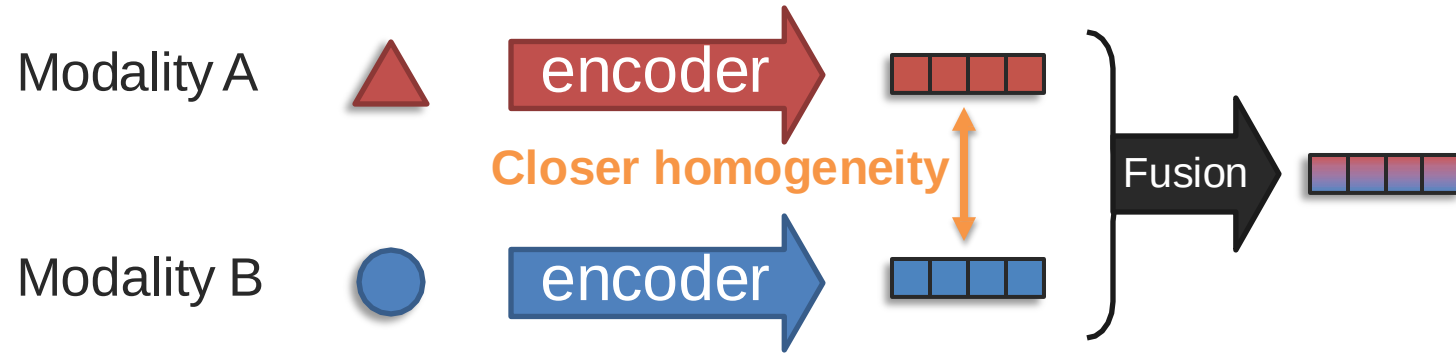
Simpler ←————→ Harder

Example: Visual Question Answering (VQA)

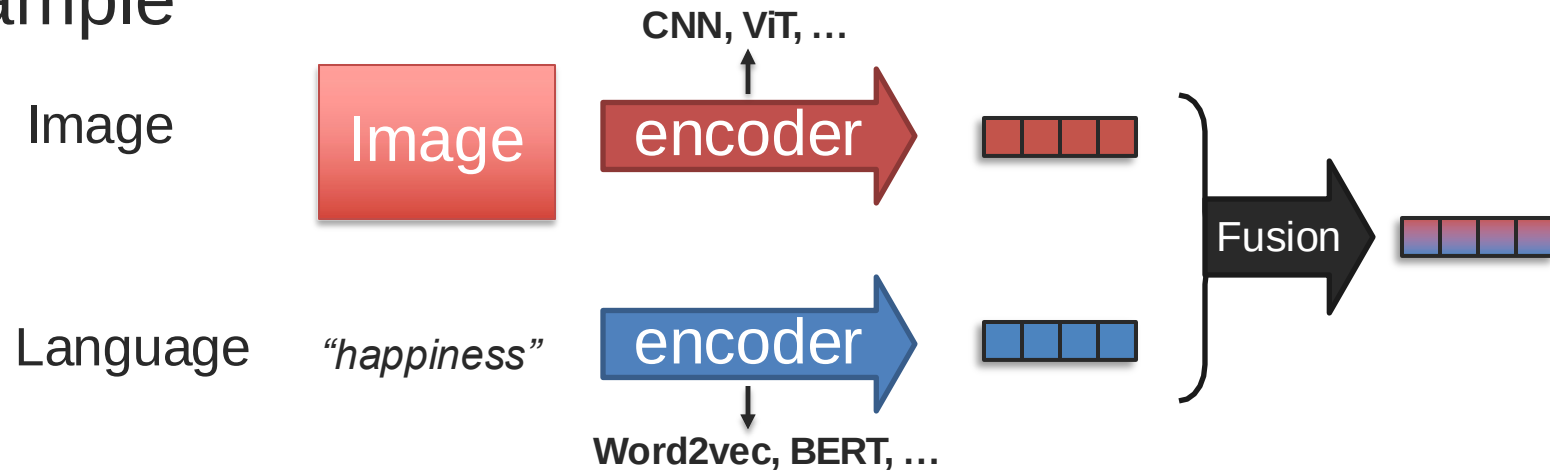


What animal is in the cage?

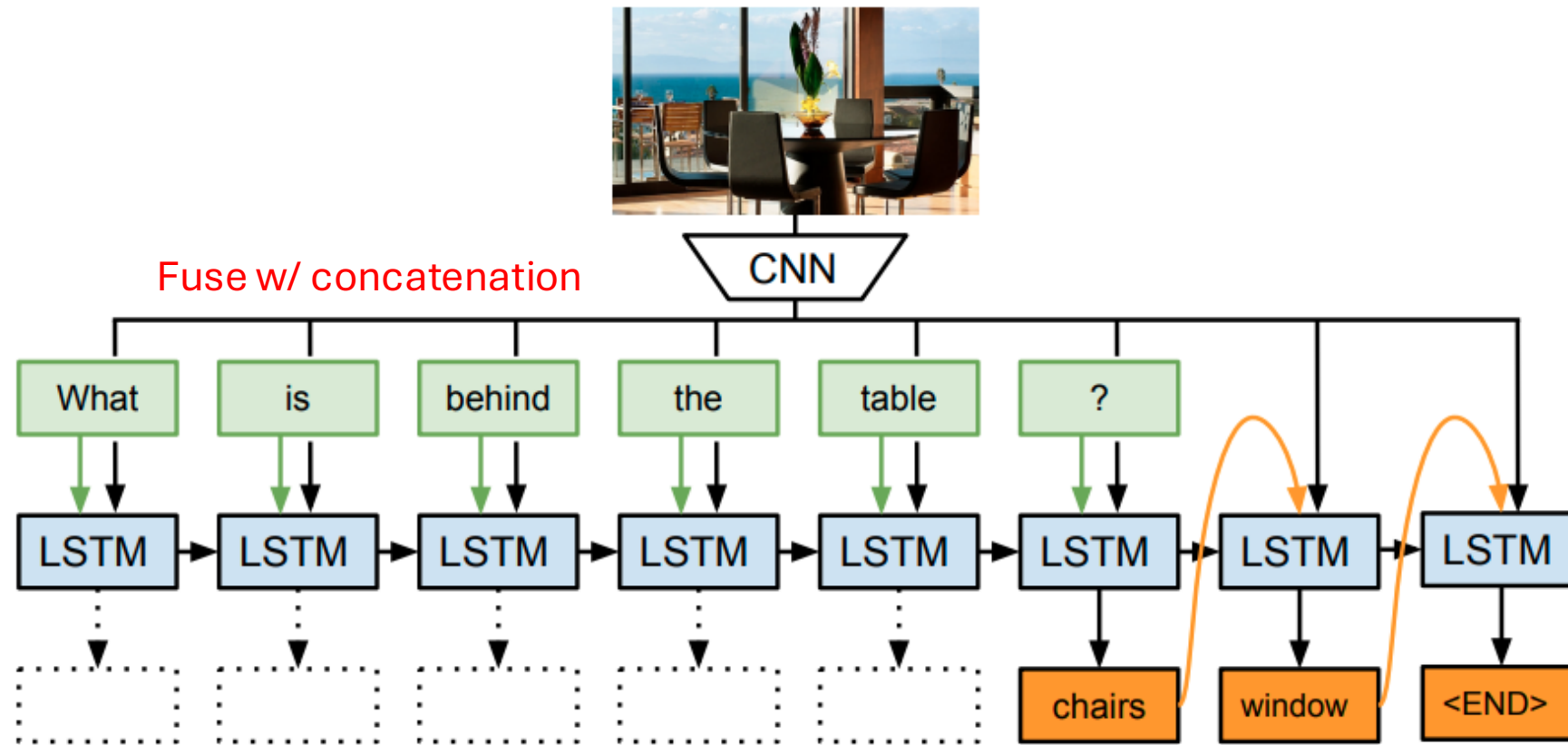
Fusion with Unimodal Encoders



Example

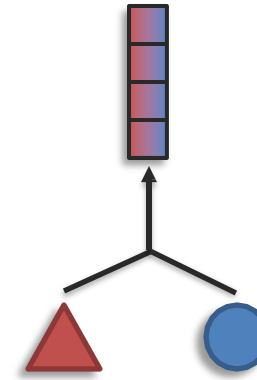


Early baseline



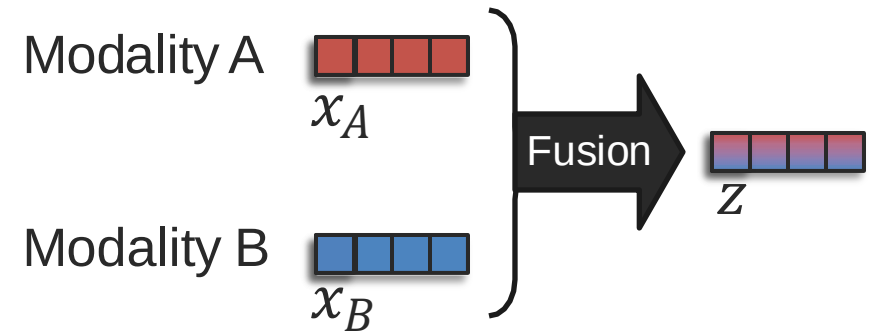
Interaction types

- Additive
- Multiplicative
- High-order Polynomial
- Gated



Interactions study

Goal: Model cross-modal interactions between the multimodal elements



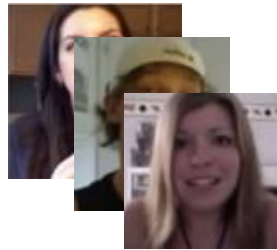
Let's study the univariate case (1-dim features) first:

$$Z = w_0 + w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

$\downarrow$ $\underbrace{\hspace{2cm}}$ $\underbrace{\hspace{2cm}}$ $\downarrow$
intercept Additive Multiplicative error
(bias term) terms term (residual term)

Interactions study: Movie reviews

300 book reviews



y : audience score (1-5)

x_A : percentage of smiling

x_B : professional status
(0=non-critic, 1=critic)

$$Z = w_0 + w_1x_A + w_2x_B + w_3(x_A \times x_B) + \epsilon$$

Diagram illustrating the components of the regression equation:

- w_0 : intercept (bias term)
- $w_1x_A + w_2x_B$: Additive terms
- $w_3(x_A \times x_B)$: Multiplicative term
- ϵ : error (residual term)

w_0 : average score when x_A and x_B are zero

w_1 : effect from x_A variable only

w_2 : effect from x_B variable only

w_3 : effect from x_A and x_B interaction only

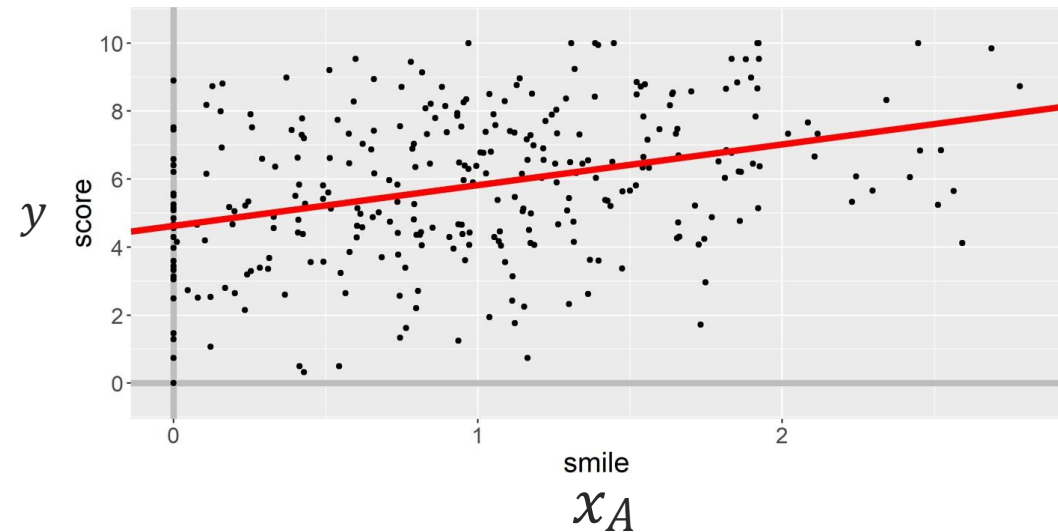
ϵ : residual not modeled by w_0 , w_1 , w_2 or w_3

Interactions study - unimodal

- *Does smiling reveal what the audience score was?*

$$z = w_0 + w_1 x_A + \epsilon$$

Estimate	
w_0	4.63
w_1	1.20

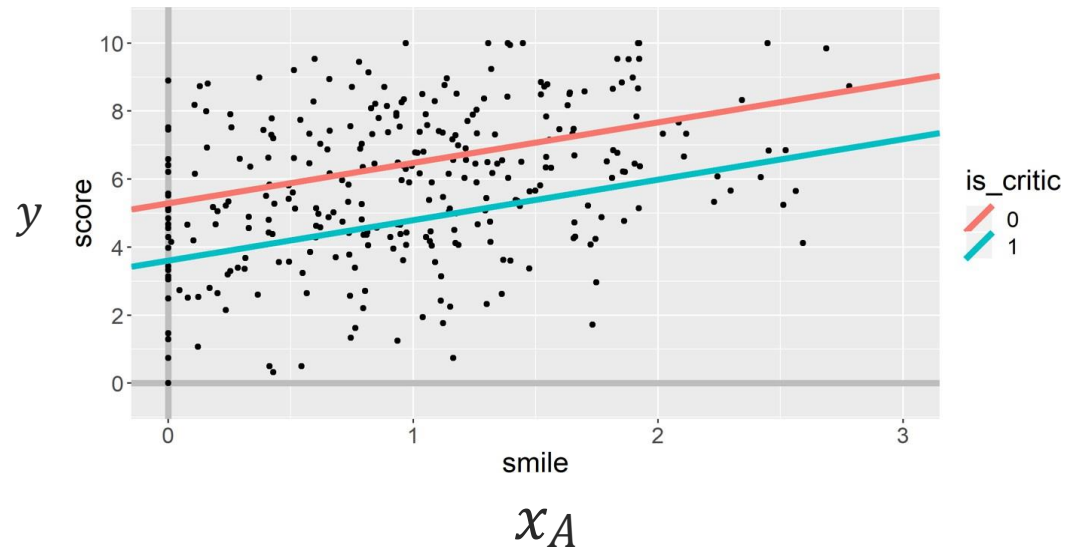


Interactions study - unimodal

- *Does the score depend on professional status?*

$$Z = w_0 + w_1 x_A + w_2 x_B + \epsilon$$

Estimate	
w_0	5.29
w_1	1.19
w_2	-1.69

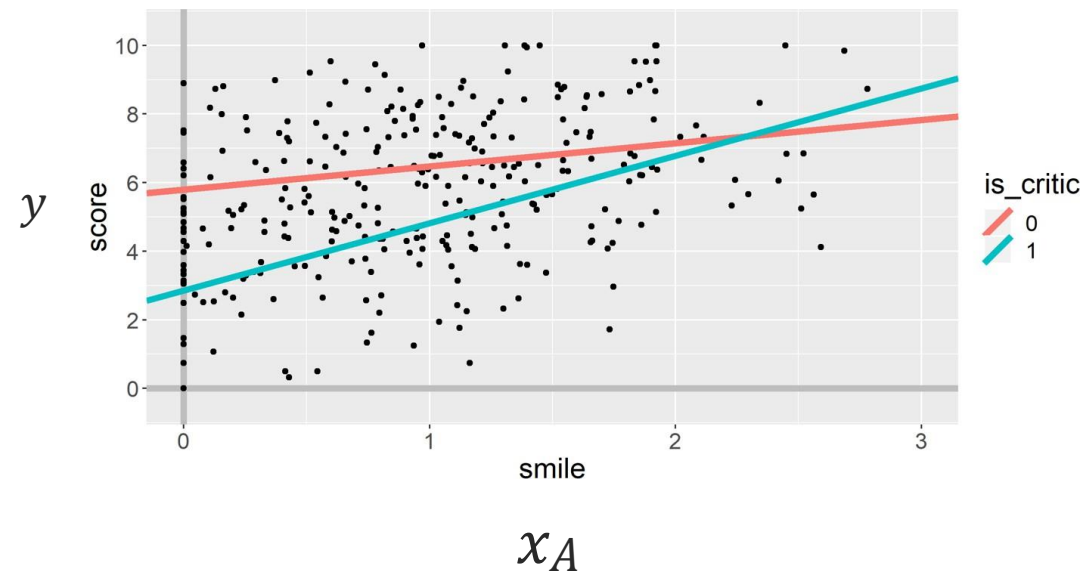


Interactions study - multiplicative interactions

Does the effect of smiling depend on professional status?

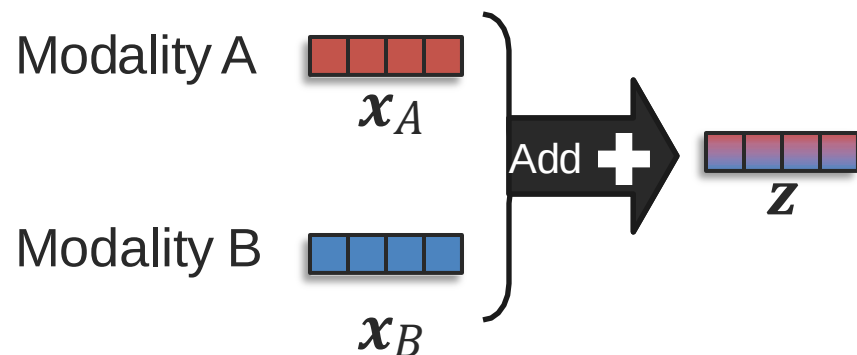
$$Z = w_0 + w_1 x_A + w_2 x_B + w_3 (x_A \times x_B) + \epsilon$$

Estimate	
w_0	5.79
w_1	0.68
w_2	-2.94
w_3	1.29

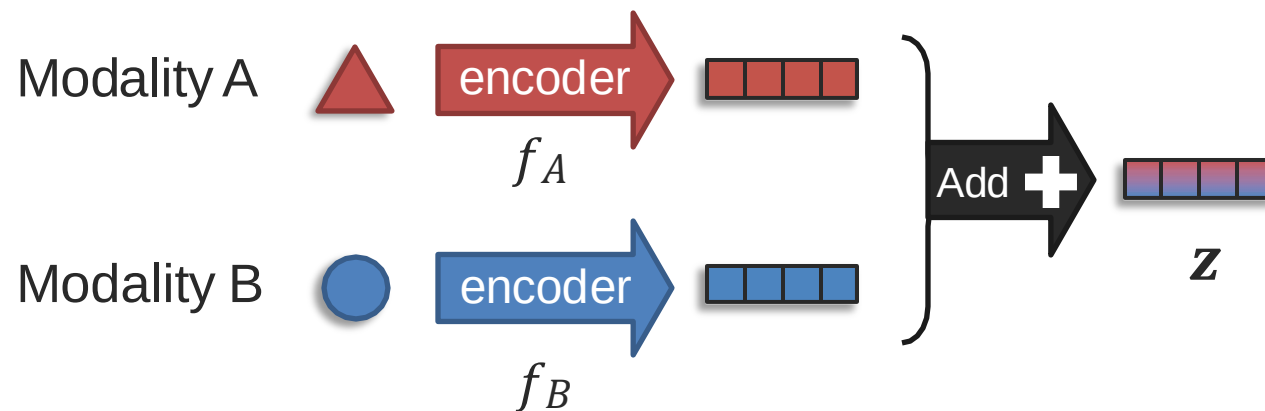


Additive Fusion: Back to multivariate case

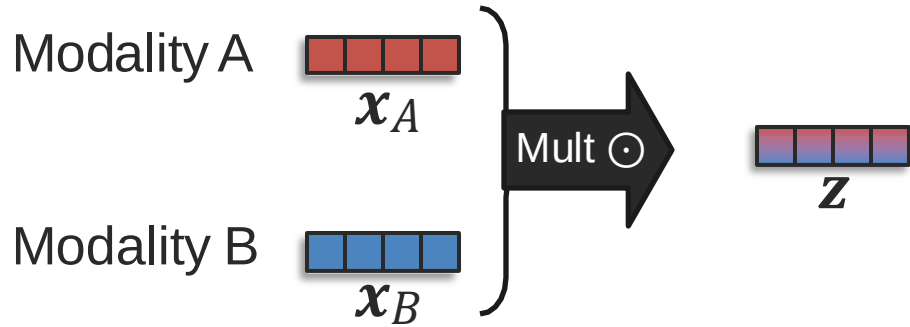
With raw features (early fusion):



With unimodal encoders (late fusion):

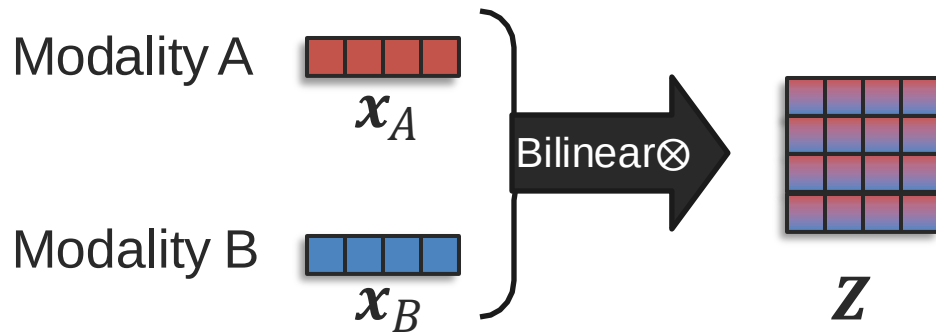


Multiplicative Fusion



Hadamard product:

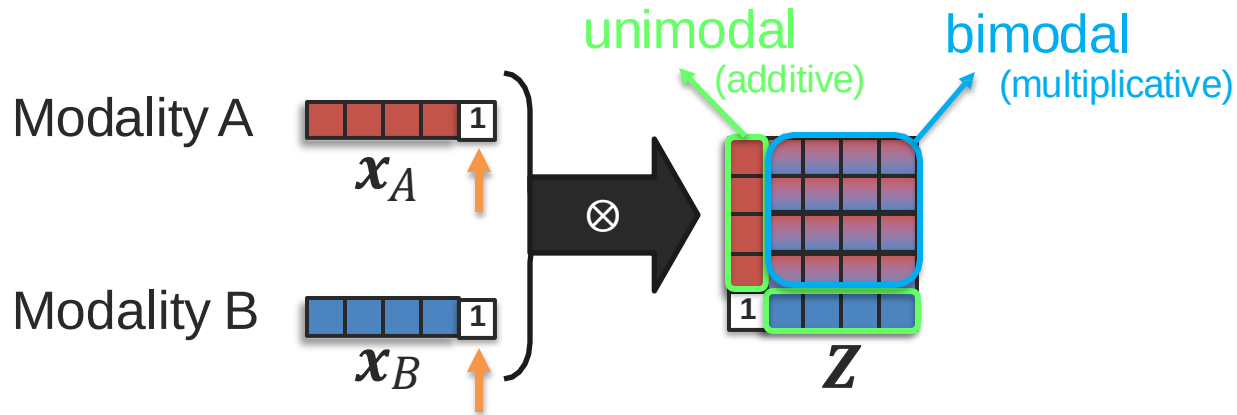
$$z = W(x_A \odot x_B)$$



Outer product:

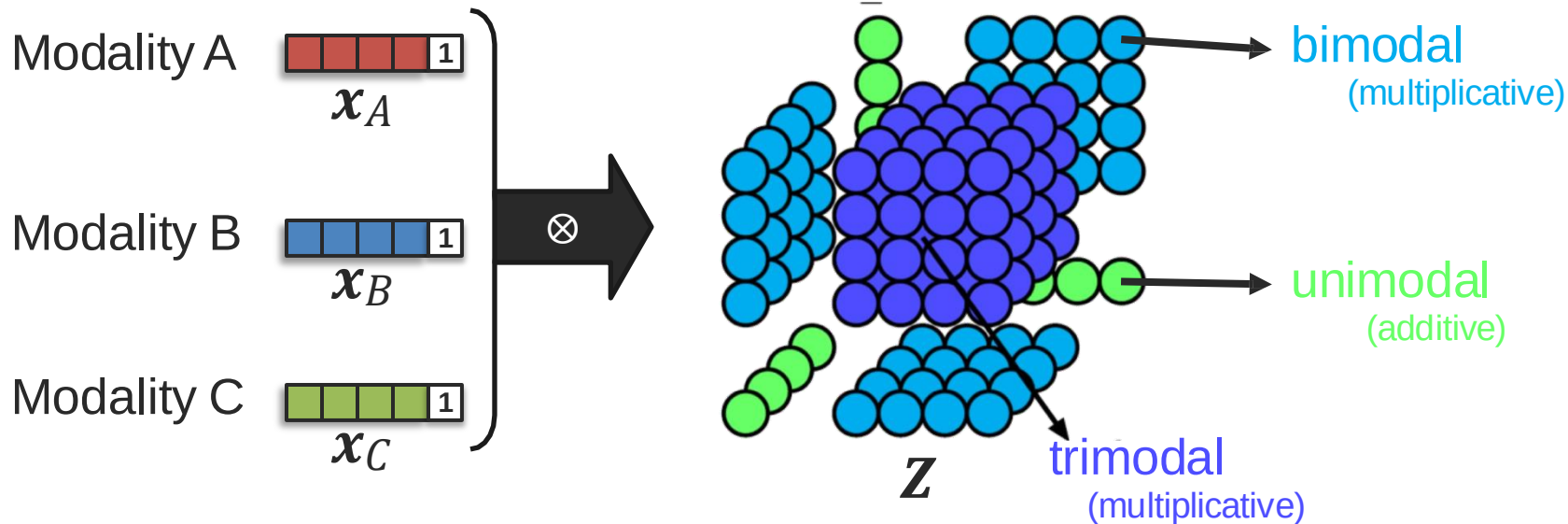
$$z = W(x_A^T \otimes x_B)$$

Tensor Fusion*



Tensor Fusion:

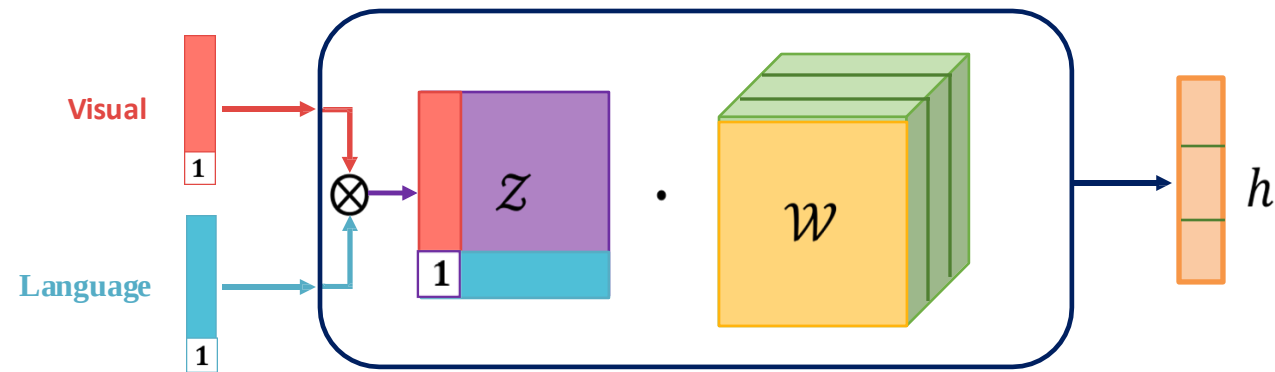
$$z = W([x_A; 1]^T \otimes [x_B; 1])$$



... but the weight matrix may end up quite large!

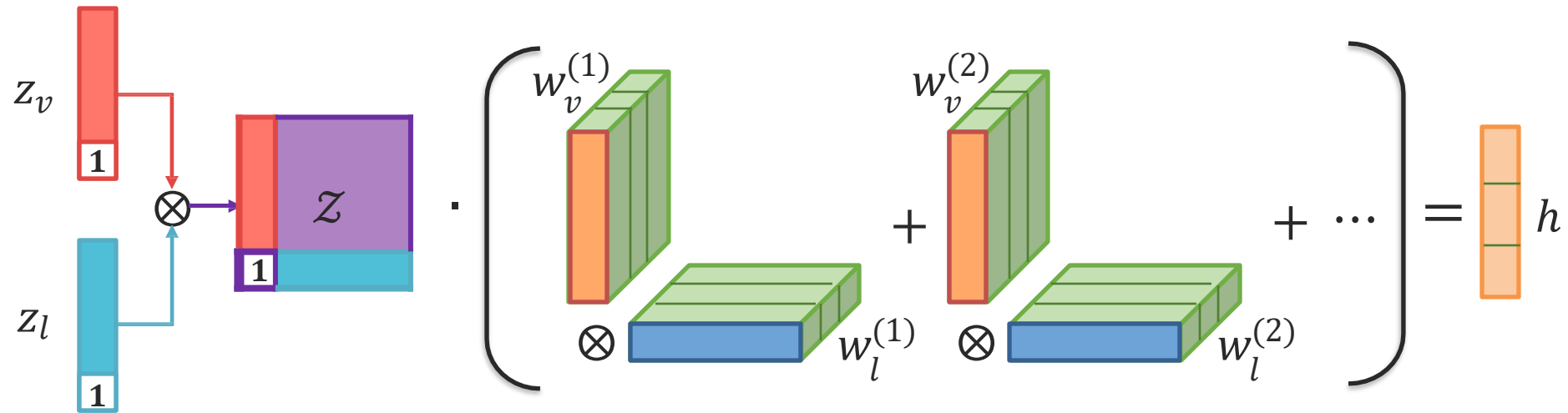
* - Zadeh et al., Tensor Fusion Network for Multimodal Sentiment Analysis, EMNLP 2017

Tensor Fusion



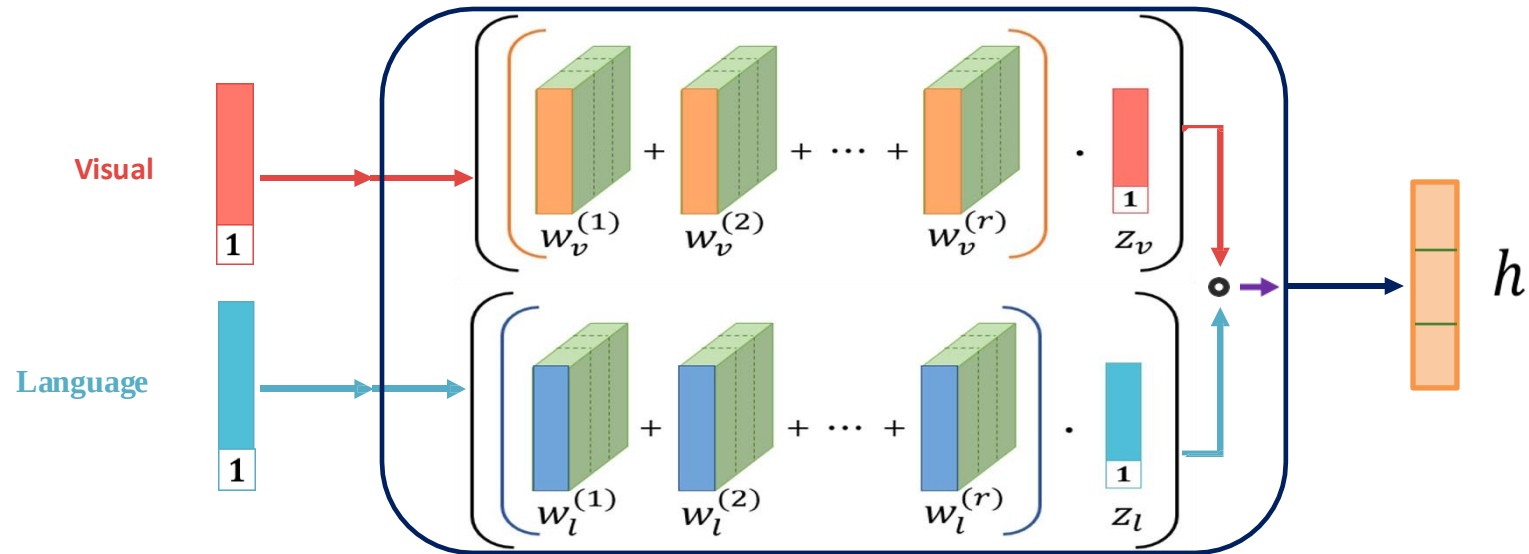
Low-rank Fusion

- Canonical Polyadic Decomposition

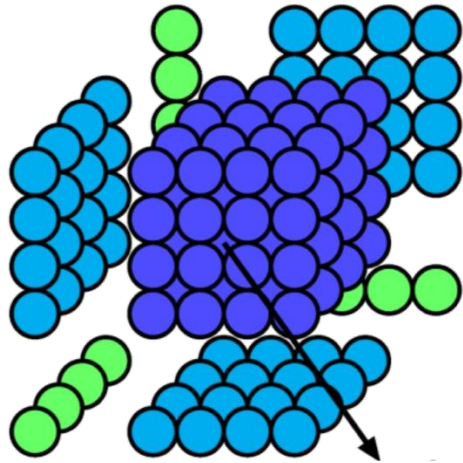


Low-rank Fusion

- Rearrange the computation



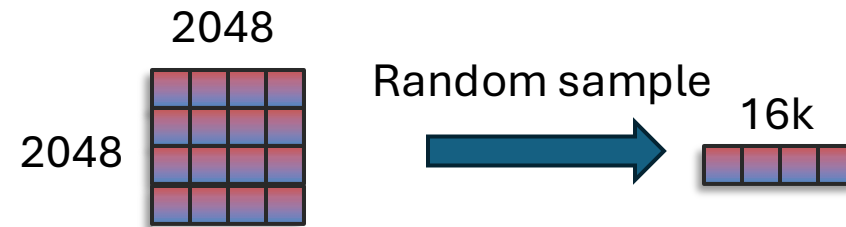
Low-rank Fusion with Trimodal Input



$$\left(\left(\begin{array}{c} \text{Low-rank factors} \\ w_v^{(1)} + w_v^{(2)} + \dots + w_v^{(r)} \end{array} \right) \cdot \begin{array}{c} 1 \\ z_v \end{array} \right) \circ \left(\begin{array}{c} \text{Low-rank factors} \\ w_l^{(1)} + w_l^{(2)} + \dots + w_l^{(r)} \end{array} \right) \cdot \begin{array}{c} 1 \\ z_l \end{array} \right) \circ \left(\begin{array}{c} \text{Low-rank factors} \\ w_a^{(1)} + w_a^{(2)} + \dots + w_a^{(r)} \end{array} \right) \cdot \begin{array}{c} 1 \\ z_a \end{array} \right)$$

Compact Bilinear Pooling

- “Sample” subset:



- Two approaches; see MCB (*):
 - Random Maclaurin (RM) [1]
 - Count Sketch [2]

* - Multimodal Compact Bilinear Pooling for Visual Question Answering and Visual Grounding; Fukui et al.

[1] - Random feature maps for dot product kernels; Kar et al.

[2] - Fast and scalable polynomial kernels via explicit feature maps; Pham et al.

Going Beyond Additive and Multiplicative Fusion?

Additive interaction:

$$z = w_1x_A + w_2x_B$$

First-order polynomial

Additive and multiplicative interaction:

$$z = w_1x_A + w_2x_B + w_3(x_A \times x_B)$$

Second-order polynomial

Trimodal fusion (e.g., tensor fusion):

$$z = w_1x_A + w_2x_B + w_3x_C + w_4(x_A \times x_C) + w_5(x_A \times x_B) + w_6(x_B \times x_C) + w_7(x_A \times x_B \times x_C)$$

Unimodal terms

(first-order)

Bimodal terms

(second-order)

Trimodal terms

(third-order)

Can we add
higher-order
interaction
terms*?

For example:

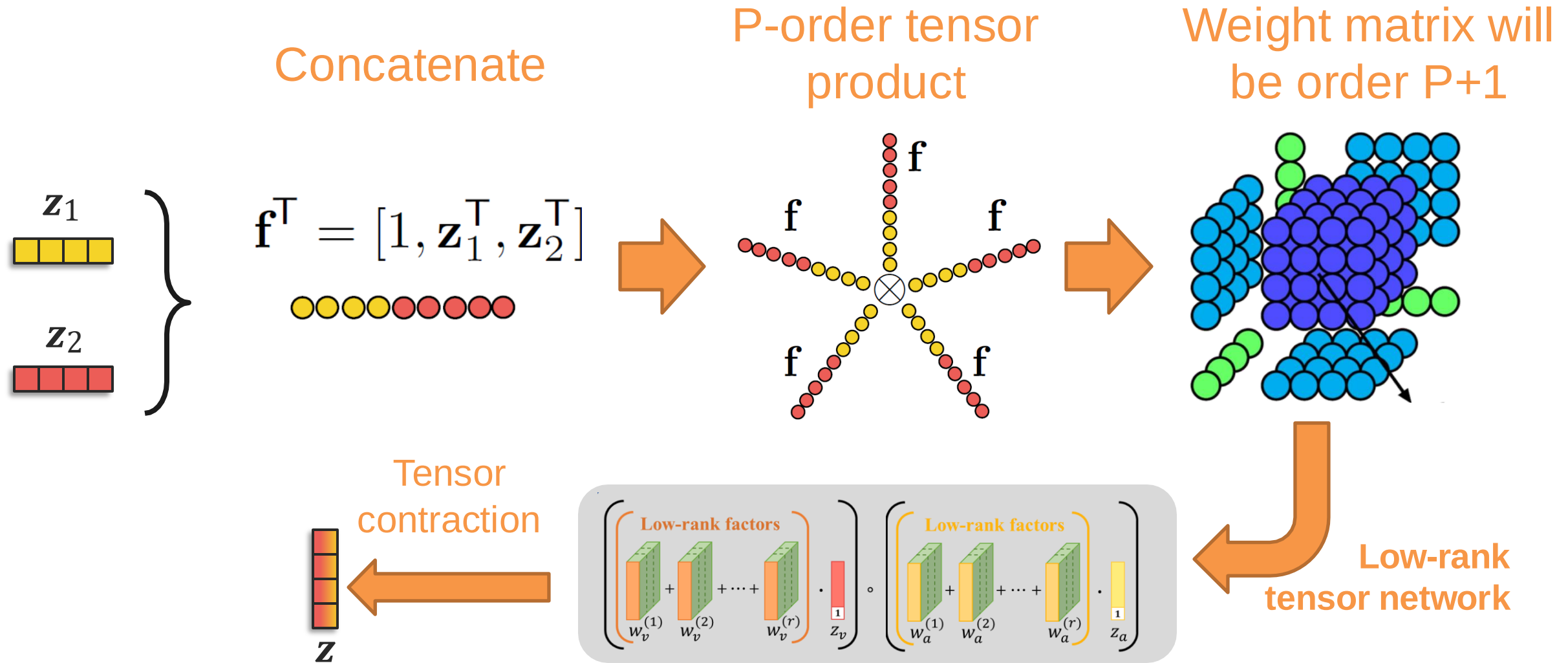
$$+w_8(x_A^2 \times x_B^2 \times x_C^2)$$

$$+w_9(x_A^3 \times x_B)$$

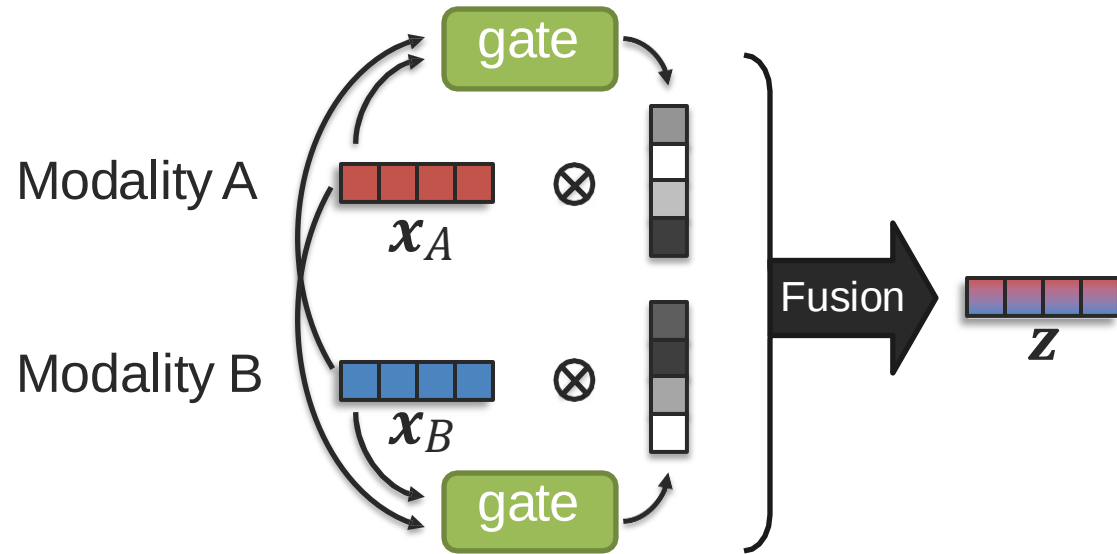
$$+w_{10}(x_B^3 \times x_C^3)$$

* - Hou et al., Deep Multimodal Multilinear Fusion with High-order Polynomial Pooling, Neurips 2019

High-Order Polynomial Fusion



Gate Module (Attentive Pooling)



Example with additive fusion:

$$z = g_a(x_A, x_B)x_a + g_b(x_A, x_B)x_b$$

g_A and g_B can be seen as attention functions

Gate Module (Attentive Pooling)



What animal is in the cage?

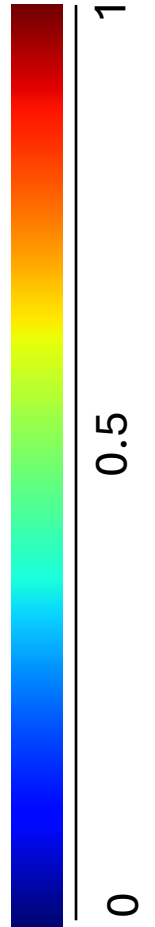
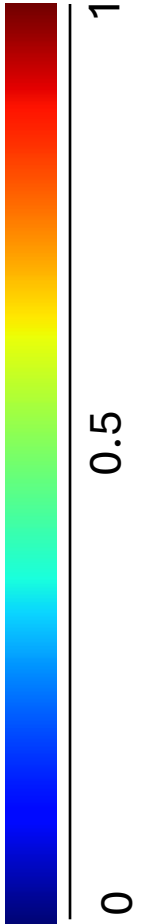


Image attention depends on the question



What color is the plastic container
inside the cage?

Word-guided attention

What is the **child** standing on ? **skateboard**



What **color** is the phone booth ? **blue**



Attention equation

- Region representation

$$a_I = \sum_{j=1..49} g_I(j) \cdot I_j$$

- Note, the image representation should have spatial dimension

3.2	5.1	9	6	...	0.1	0.4	7.9
-----	-----	---	---	-----	-----	-----	-----

I_j

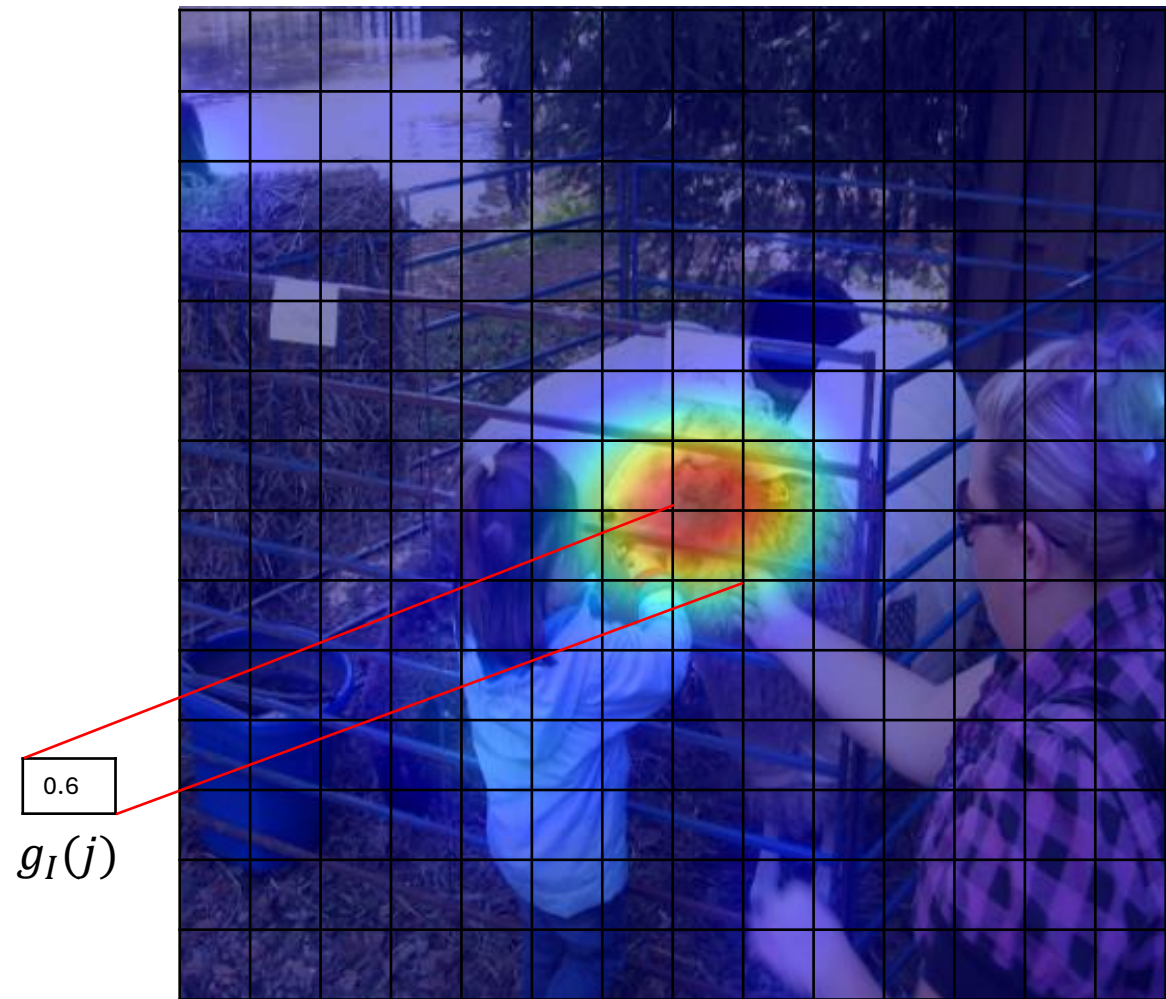


Attention gate equation

- Image attention:

$$a_I = \sum_{j=1..49} g_I(j) \cdot I_j$$

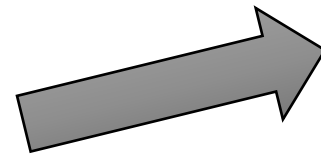
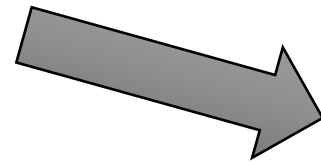
- How to compute g_I ?



Correlation matrix



What animal
is in the cage?



C

			
What	1.2	2.3	1.7
animal	-0.4	6.5	4.2
is	1.3	1.2	0.6

Word-guided attention

- Marginalize with max

			
What	1.2	2.3	1.7
animal	-0.4	6.5	4.2
is	1.3	1.2	0.6



Max
→

	1.3
	6.5
	4.2

Softmax
→

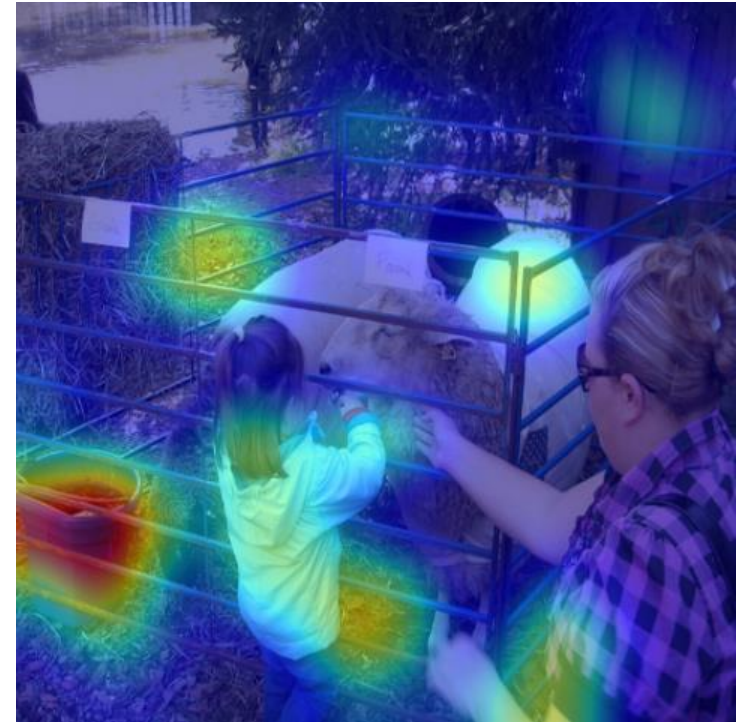
0.005
0.904
0.091

Gate visualization

- Reacts to highly word-correlated regions



What animal is in the cage?



What color is the plastic container inside the cage?

Gate attention works well for alignment (i.e., detect a word)



Can you see cars?



Are there any buildings?



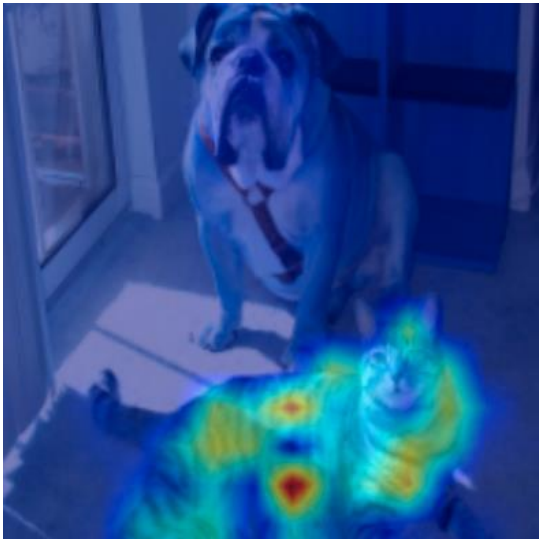
Are there trees?

Model explainability

- Attention for global signal (given the input)
- Gradients for local signal (given a prediction)

The prediction:

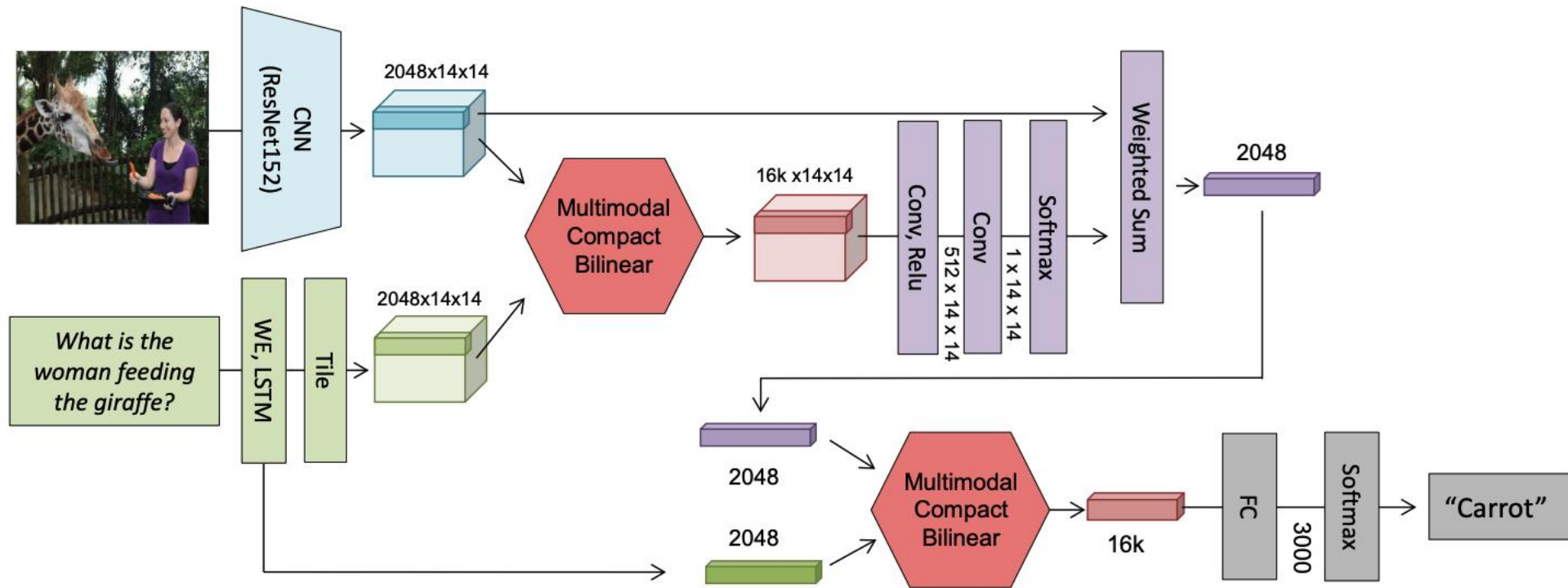
Cat



Dog



State-of-the-art for VQA w\ MCB (2019)

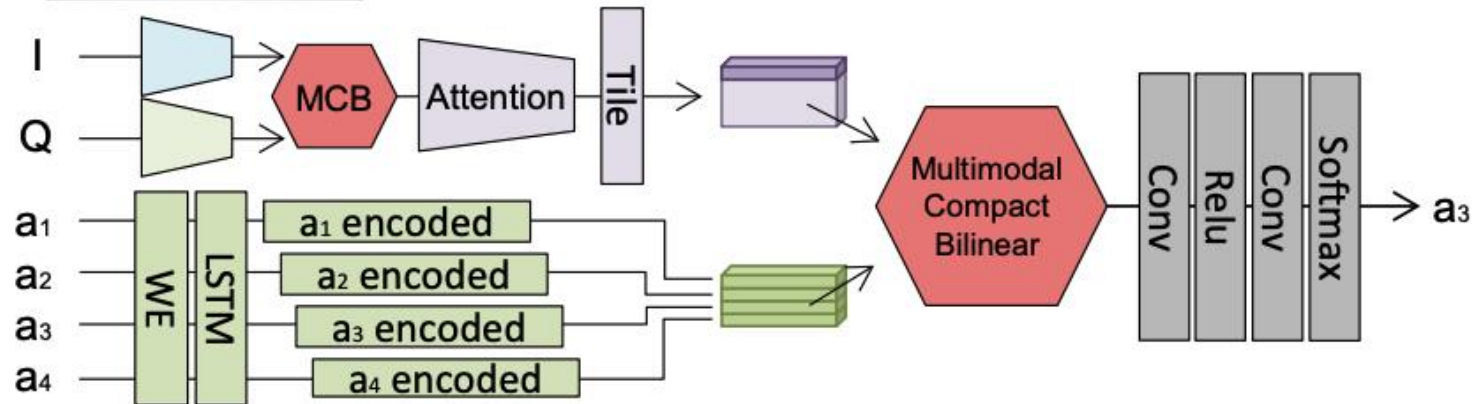


* - Multimodal Compact Bilinear Pooling for Visual Question Answering and Visual Grounding

Trimodal with MCB



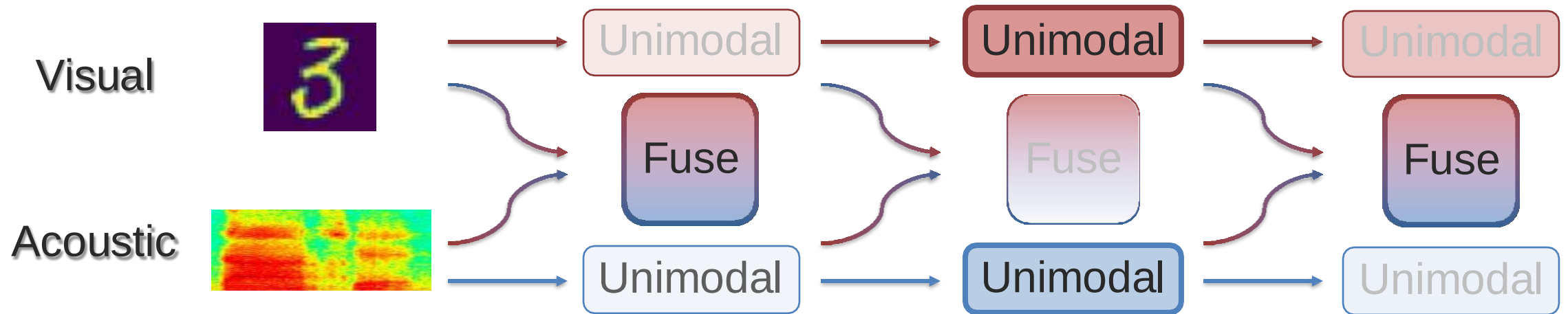
Q : "What do you see?" (Ground Truth : a_3)
 a_1 : "A courtyard with flowers"
 a_2 : "A restaurant kitchen"
 a_3 : "A family with a stroller, tables for dining"
 a_4 : "People waiting on a train"



Ablation study (taken from the MCB paper)

Method	Accuracy
Element-wise Sum	56.50
Concatenation	57.49
Concatenation + FC	58.40
Concatenation + FC + FC	57.10
Element-wise Product	58.57
Element-wise Product + FC	56.44
Element-wise Product + FC + FC	57.88
MCB ($2048 \times 2048 \rightarrow 16K$)	59.83

When to fuse?



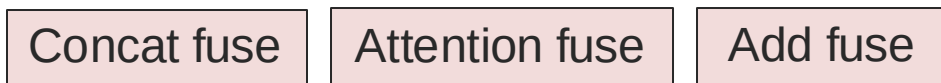
Dynamic Early Fusion

Fusion fully learned from optimization and data

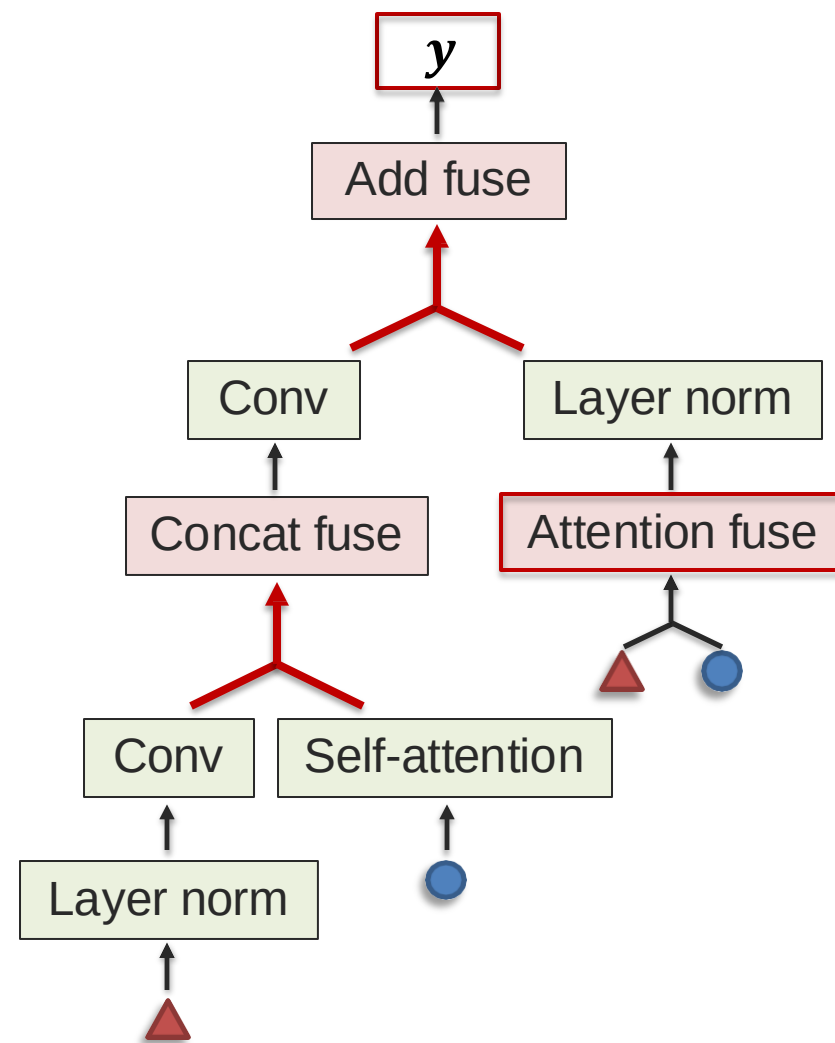
1. Define basic representation building blocks



2. Define basic fusion building blocks



3. Automatically search for composition using neural architecture search



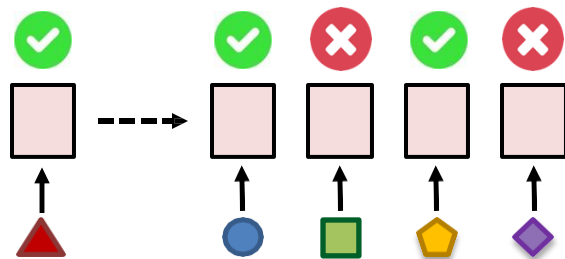
[Xu et al., MUFASA: Multimodal Fusion Architecture Search for Electronic Health Records. AAAI 2021]

[Liu et al., DARTS: Differentiable Architecture Search. ICLR 2019]

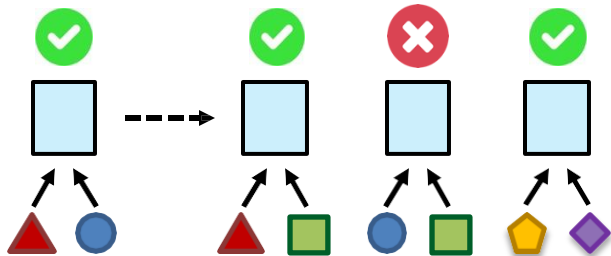
Heterogeneity-aware Fusion

Information transfer, transfer learning perspective

1a. Estimate modality heterogeneity via transfer



1b. Estimate interaction heterogeneity via transfer



2a. Compute modality heterogeneity matrix

	△	●	■	⬠	⬡
△	0				
●	1	0			
■	3	2	0		
⬠	1	2	3	0	
⬡	5	4	6	3	0

2b. Compute interaction heterogeneity matrix

	{△, ●}	{△, ■}	{●, ■}	{⬠, ⬡}
{△, ●}	0			
{△, ■}	1	0		
{●, ■}	3	2	0	
{⬠, ⬡}	1	2	4	0

3. Determine parameter clustering

$$\mathbb{U}_1 = \{U_1, U_2, U_4\}$$

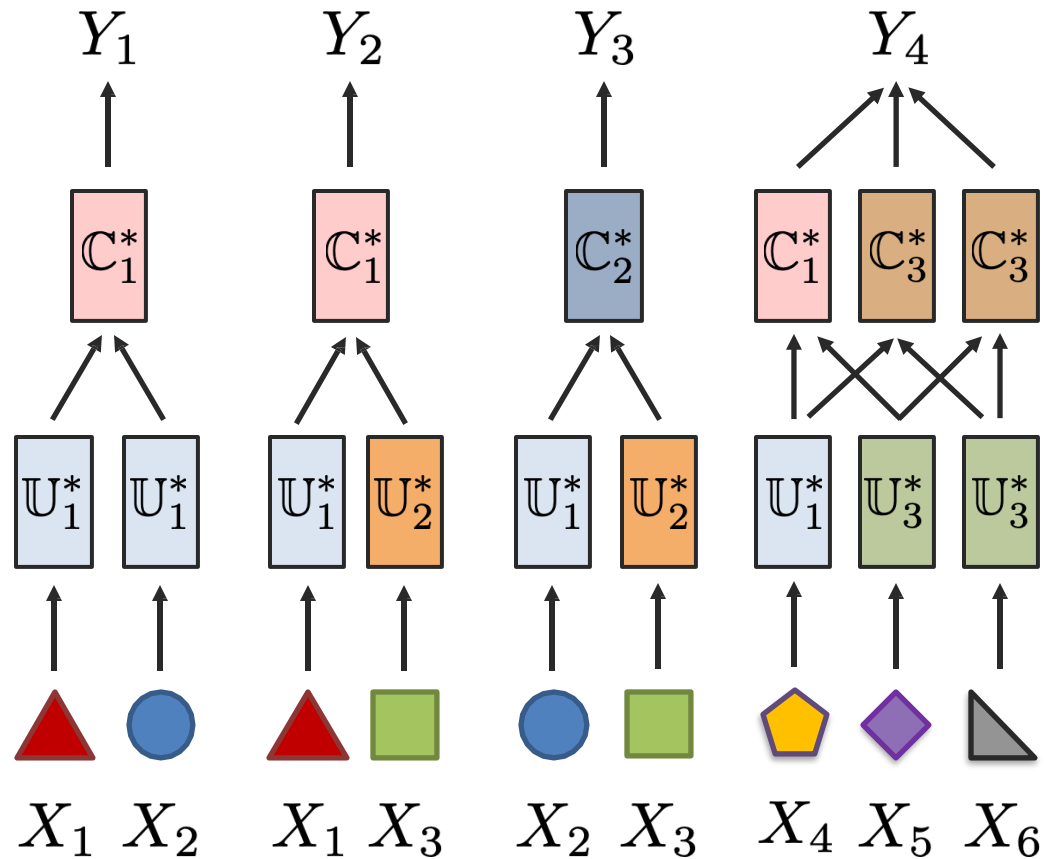
$$\mathbb{U}_2 = \{U_3\}$$

$$\mathbb{U}_3 = \{U_5\}$$

$$\mathbb{C}_1 = \{C_{12}, C_{13}, C_{45}\}$$

$$\mathbb{C}_2 = \{C_{23}\}$$

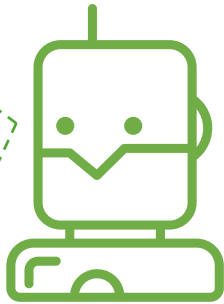
Heterogeneity-aware Fusion



Evaluating Cross-Modal Learning



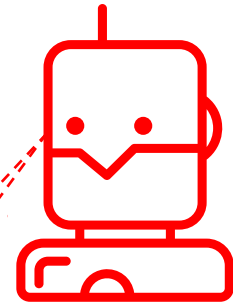
Q: How many animals are pictured?



Percives the whole input



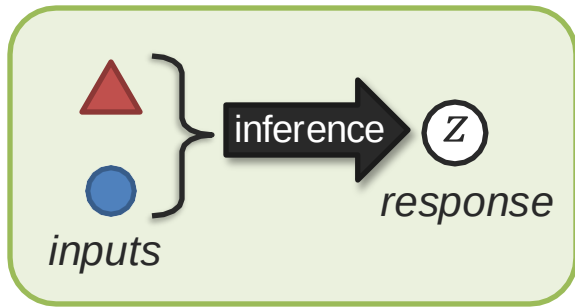
Q: How many animals are pictured?



Perceives only the question

Suppose the answer is “2”, how can you tell if the model used cross-modal interactions ?

Recall: Taxonomy of Interaction Responses

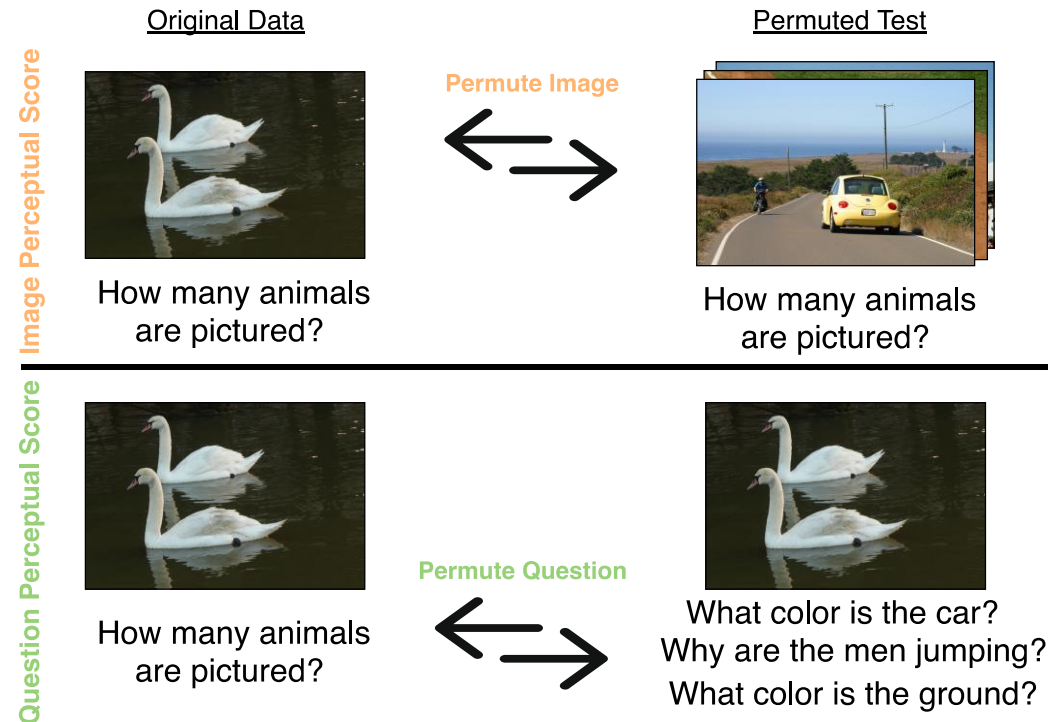


	signal	response	
Redundancy	a.		Equivalence
	b.		
Nonredundancy	a+b		Enhancement
	a+b		
	a+b	 and 	Independence
	a+b		Dominance
	a+b	 (or )	Modulation
	a+b		Emergence

Not available

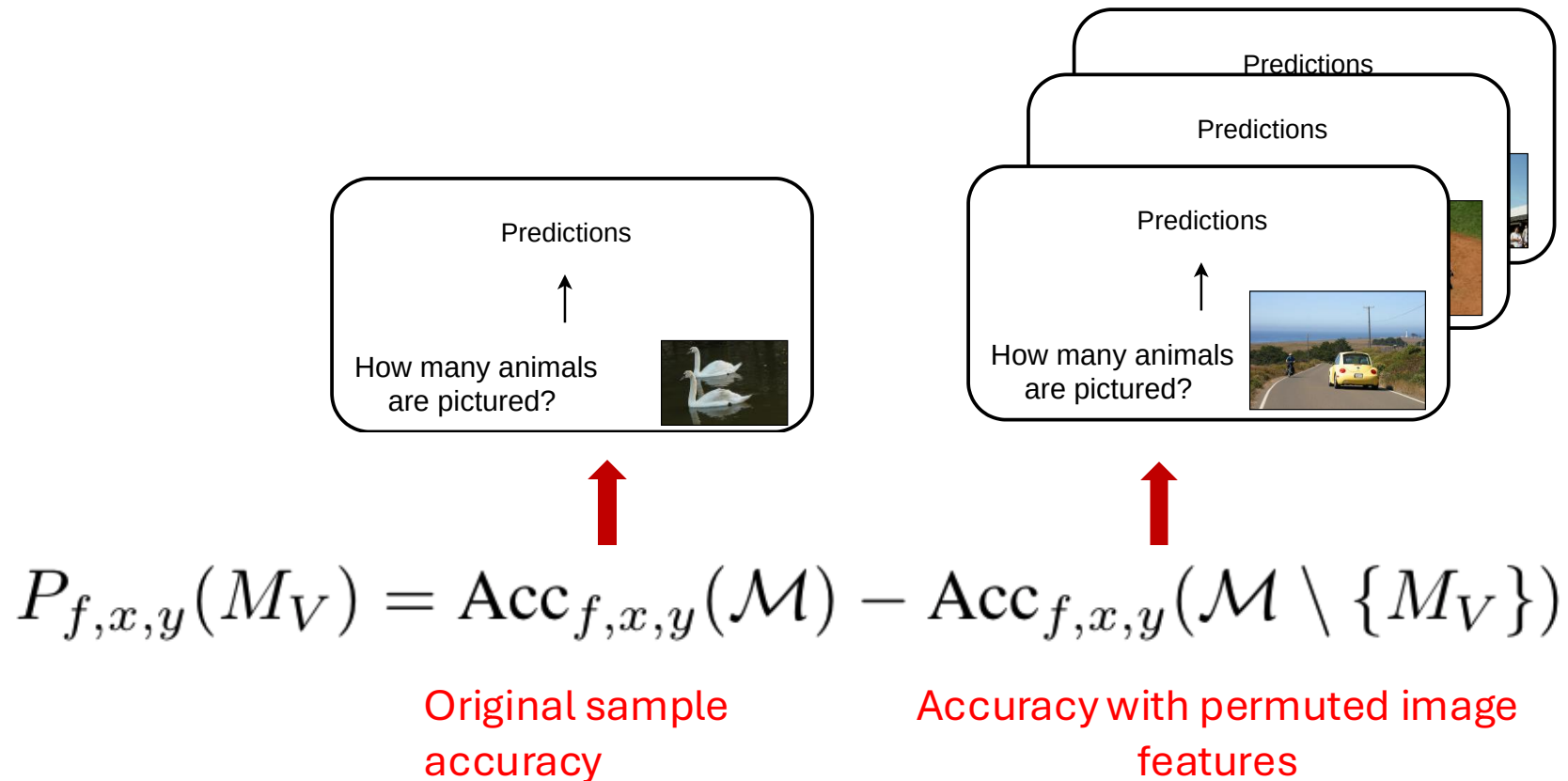
Perceptual Score^[1]

- We generate a permuted test set

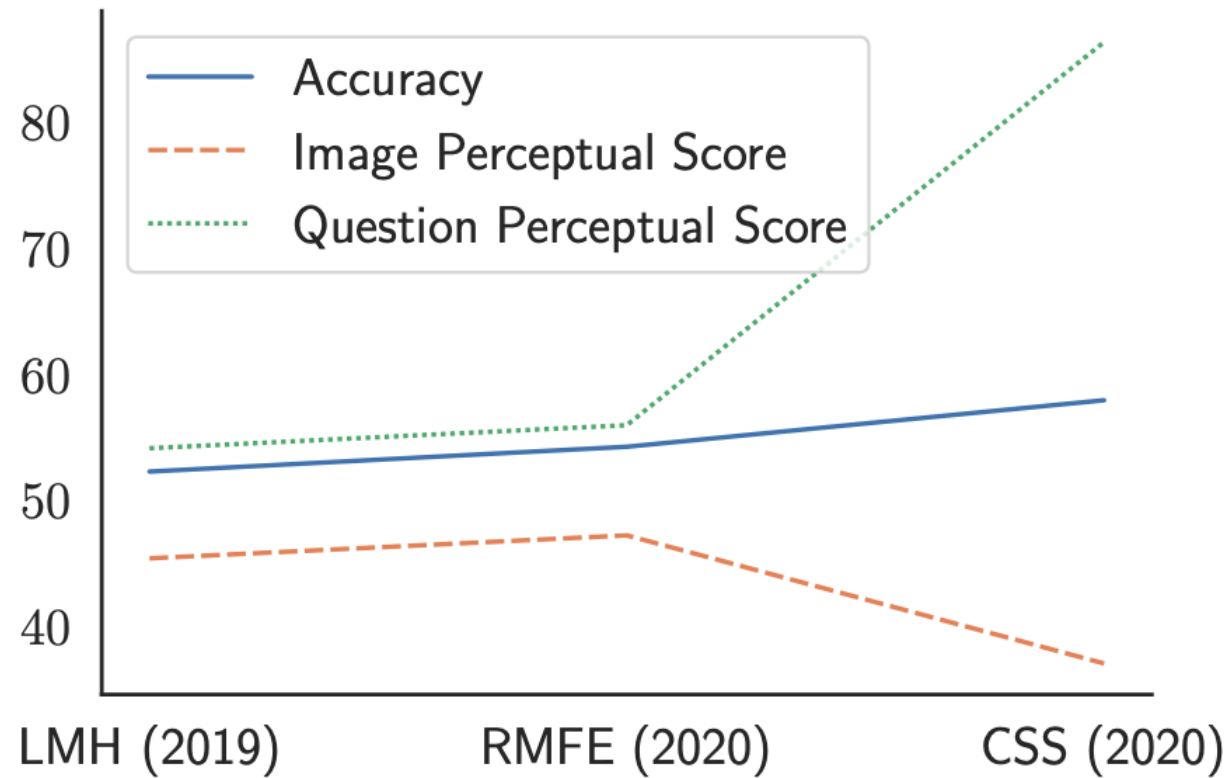


Sample Perceptual Score

A set of input modalities $\mathcal{M} = (M_V, M_Q)$



Progress of VQA-CP models



Why?

- “how many” questions are typically answered with 2



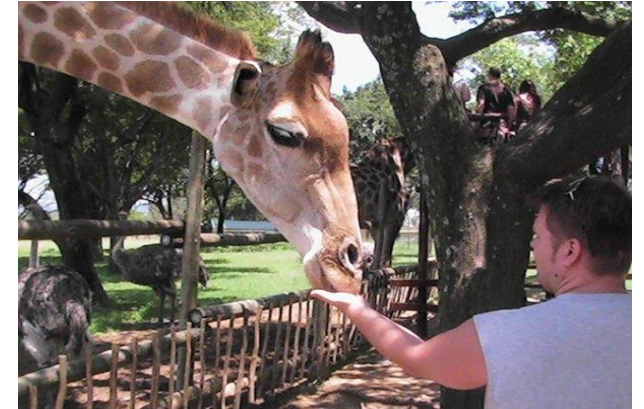
Q: How many flags are there?

A: 2



Q: How many levels are on this bus?

A: 2



Q: How many different type of animals are there?

A: 2

Bad Cross-Modal Learning leads to Poor Generalization



Q: How many people are in the image

A: 4



Q: How many buses are there?

A: 4

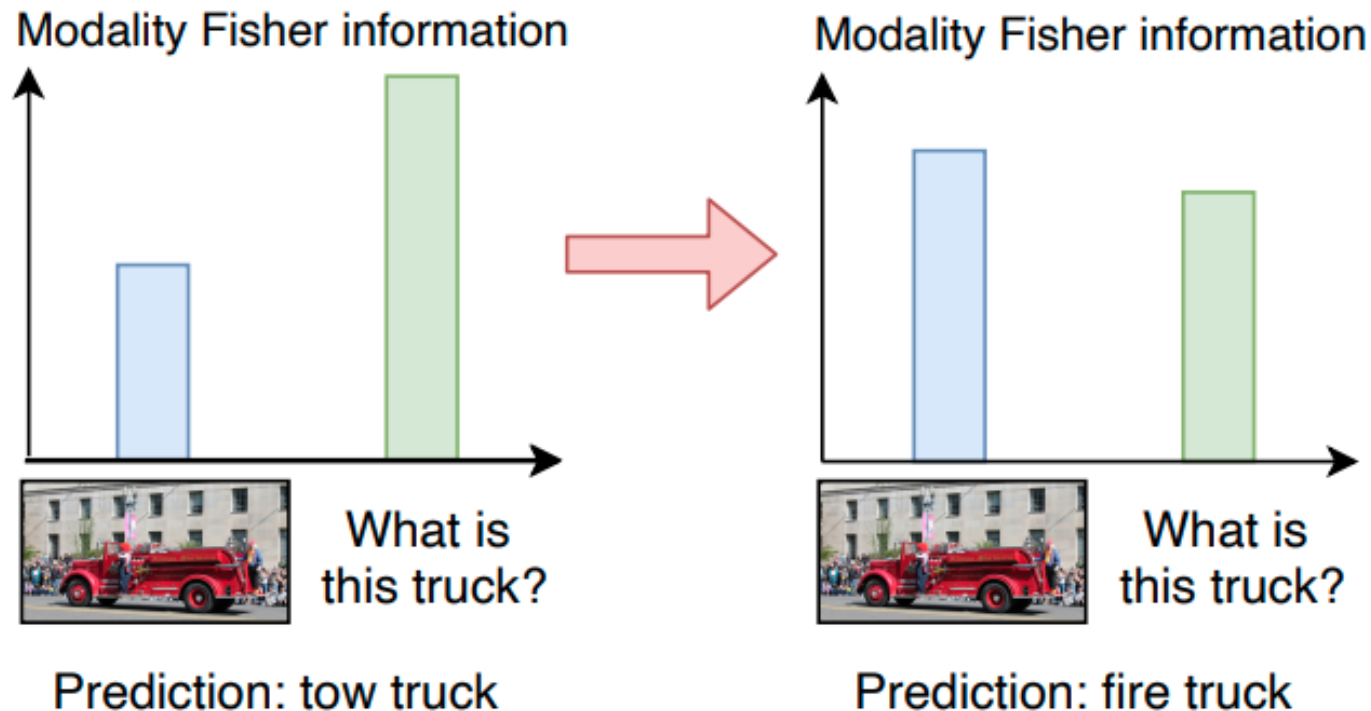


Q: How many bikes are there?

A: 6

Improve perceptivness: Maximize fisher information^[2]

A regularization maximize information from each modality



The Greedy Nature of Learning in Multi-modal

Kinetics dataset



(a) headbanging



(c) shaking hands



(e) robot dancing



(g) riding a bike



Adding more modalities should always help?

Modalities: RGB (video clips)

A (Audio features)

OF (optical flow - motion)

Dataset	Multi-modal	V@1	Best Uni	V@1	Drop
Kinetics	A + RGB	71.4	RGB	72.6	-1.2
	RGB + OF	71.3	RGB	72.6	-1.3
	A + OF	58.3	OF	62.1	-3.8
	A + RGB + OF	70.0	RGB	72.6	-2.6

But sometimes multimodal doesn't help! **Why?**

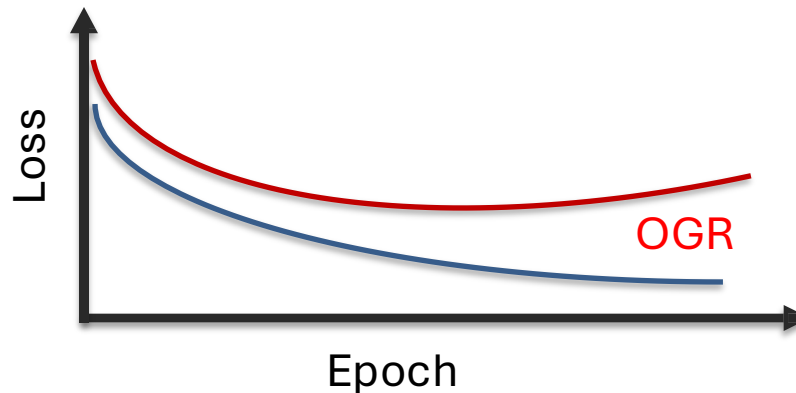
[Wang et al., What Makes Training Multi-modal Classification Networks Hard? CVPR 2020]

[Wu et al., Characterizing and Overcoming the Greedy Nature of Learning in Multi-modal Deep Neural Networks. ICML 2022]

Multimodal overfitting

1. Multimodal networks are more prone to overfitting due to increased complexity
2. Different modalities overfit and generalize at different rates

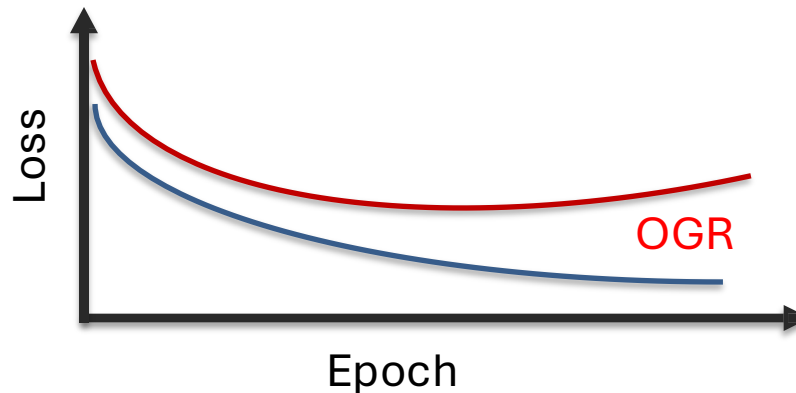
We can calculate the overfitting-to- generalization ratio per modality (OGR)



Multimodal overfitting

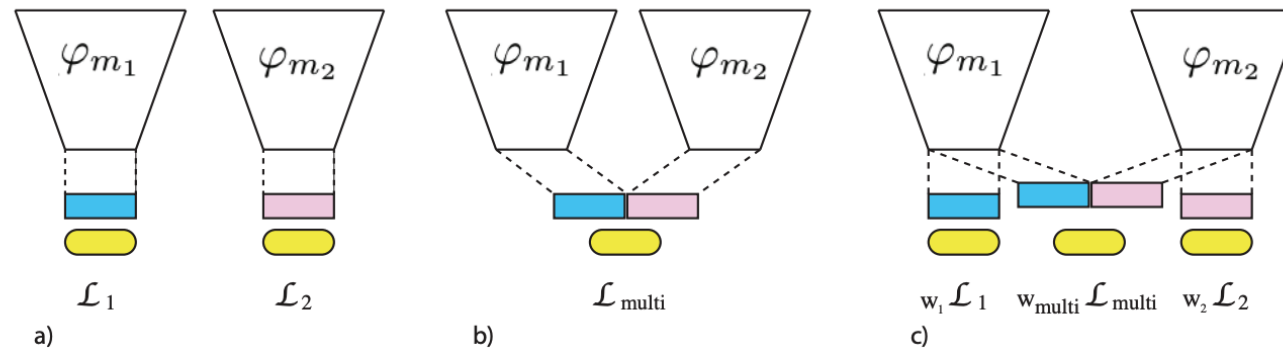
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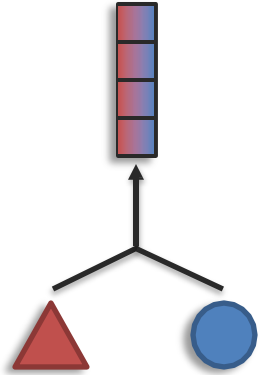


Improve optimization

- Have a loss per modality
- Simultaneously train unimodal networks to estimate OGR for each modality
- Reweight multimodal loss using unimodal OGR values (Gradient blending)

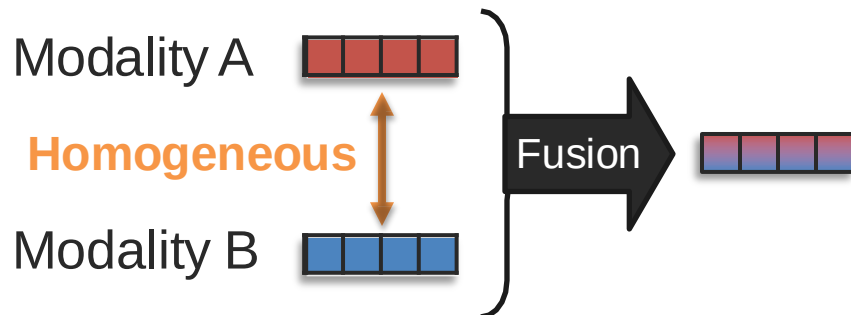


Representation Fusion

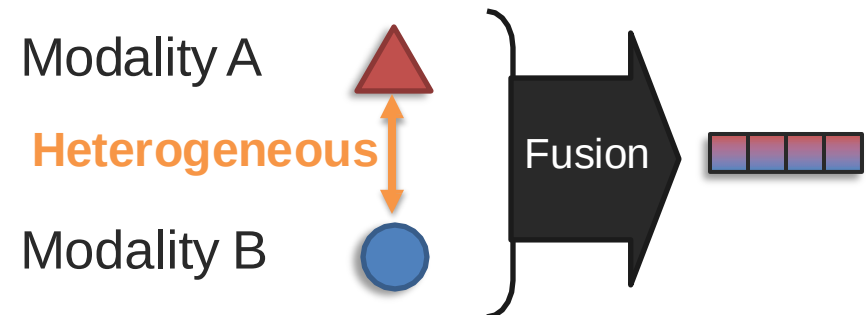


Definition: Learn a joint representation that models cross-modal interactions between individual elements of different modalities

Basic fusion:



Raw-modality fusion:



Simpler ←————→ Harder

Summary

